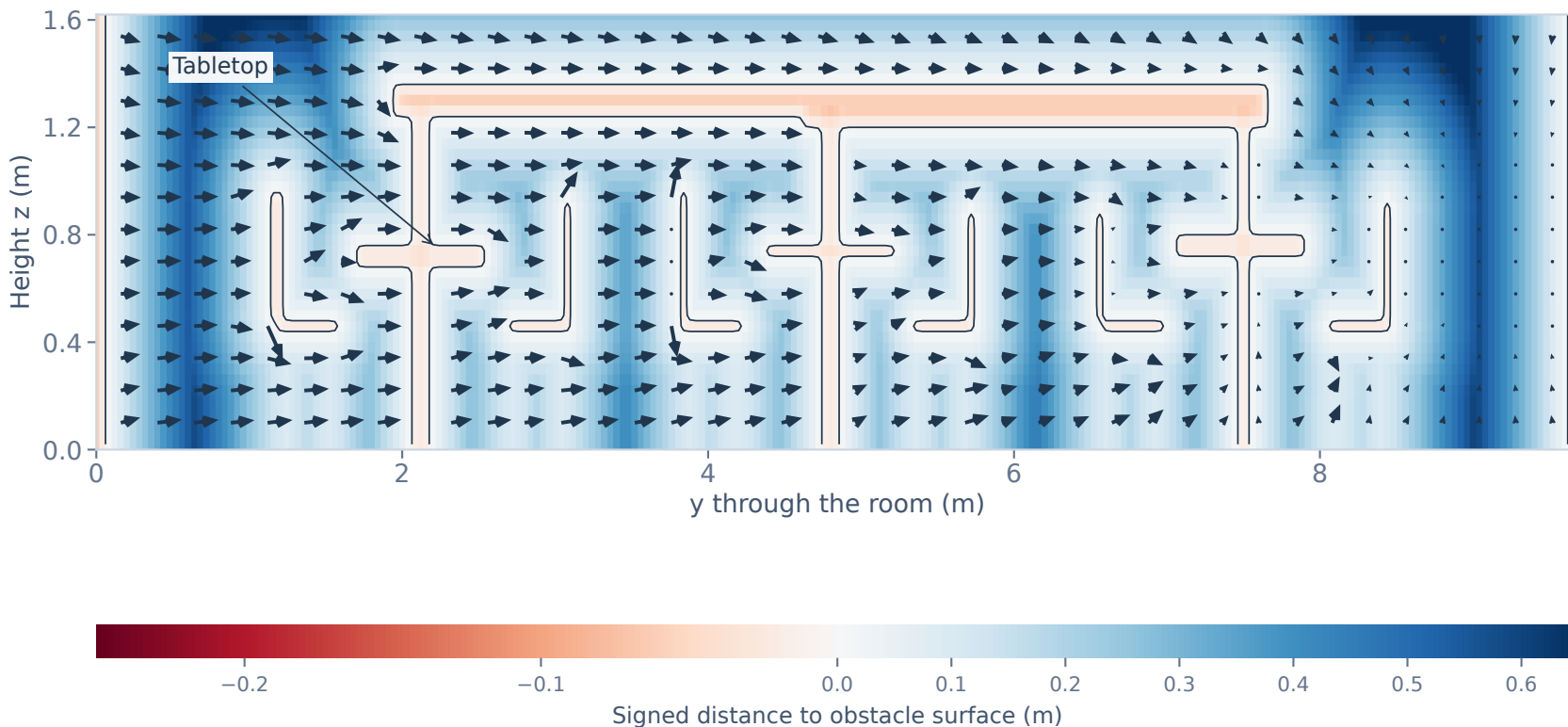


Why the furniture field is three-dimensional

Actual stored SDF and guidance vectors through the center table column: $x = 4.48$ m

Signed distance + guidance projected into this vertical slice



Blue: free space. Red: obstacle interior. Arrows are stored field values projected into the y - z plane; the full guidance also has an x component. This is a field visualization of the current training geometry, not a learned rollout or a collision-free traversal certificate.

01 Occupancy

Canonical furniture boxes
voxelized on a 4 cm grid.

02 CAT potential fields

Original SDF $\rightarrow$ boundary gradient
 $\rightarrow$ progressive 3D fast marching.

03 Whole-body queries

Hands and body sample clearance
and guidance in 3D.

39 scenes in one bank

36 byte-verified original scenes
+ 1 explicit reconstruction
+ 2 dense FMM clutter scenes